Leon Sandler
Independent Researcher
Northbrook, Illinois, USA
sandler.leon@gmail.com
ORCID 0009-0007-4584-808X

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The Editor-in-Chief
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Dear Editor,

I submit the manuscript “Probabilistic Design and Safety-Constrained Optimization of a Firewater-Powered Portable Valve Actuation System” as a Technical Paper for consideration in the Journal of the Brazilian Society of Mechanical Sciences and Engineering.

Large valves must sometimes be operated when plant power, instrument air and hydraulic supply are lost. Almost every hazardous facility already owns a pressurized, regularly inspected source of hydraulic power, its firewater system, and the manuscript studies a portable water-hydraulic valve operator driven through a fire hose. It does not argue for a product. It treats the tool as a computational engineering problem with three conflicting requirements, completing the stroke, not exceeding the allowable stem torque, and not taking the water that firefighting needs, and asks which design and operating rules satisfy them, and how far the answer can be trusted.

The manuscript makes four contributions. (i) A probabilistic model with an explicit hydraulic circuit (source characteristic, hose, motor leakage and start-up efficiency, relief valve, and the pressure surge on stall), in which each added mechanism can be switched off so that the earlier first-order model is recovered exactly. (ii) Separate safety criteria: a safe stroke, exceedance of the allowable stem torque, stem damage and violation of the firefighting reserve. The model shows that the first-order estimate of about 80% success was optimistic: a safe stroke occurs in 63% of the draws for the baseline tool, with an over-torque probability of 15%. (iii) A safety-constrained multi-objective design study, whose optimum is re-evaluated out of sample and under deliberately wrong input distributions: with a hose-side accumulator and a relief chart set from the stem strength, the over-torque probability can be brought to 1% at a cost of 3 to 9 percentage points of safe strokes, a cost that a slow creep flow at the start and at the end of travel recovers in part (4 to 6 points), but the optimum does not survive a stem 30% weaker than its data sheet (over-torque 14%), and the manuscript says so. (iv) A reconciliation of the water budget: a stroke uses about 1,600 L of water, not the 50 to 100 L/min first assumed.

The study is explicit about what it is. Every input distribution is an assumed prior, because no published distribution of breakaway torques, surge parameters or spare firewater flows was found; the model is verified numerically (draw-for-draw equivalence with the earlier model, closed-form checks, energy balance, 31 automated tests) but not validated physically; and the results are simulation outputs under those assumptions, not measurements. The surge relations are checked against classical water-hammer

results and a method-of-characteristics simulation (conservative in 97% of random cases, provided the relief valve responds within about 0.1 s), the over-torque probabilities of the chosen designs are given with confidence limits and a strict compliance rule, and preliminary mechanical loads (gearbox rating, reaction arm, accumulator size) are estimated. The manuscript lists the measurements that would replace the assumptions, and it makes no claim of superiority over conventional operators.

The code and results are openly available at https://github.com/sandlerleon/hydratorque-firewater-valve-actuation-mc and archived at https://doi.org/10.5281/zenodo.23969346 (code, all versions); the manuscript and its Supplementary Information are archived at https://doi.org/10.5281/zenodo.23969344 (all versions). One command regenerates every number and figure.

Disclosure. The concept originated as a white paper that I prepared for an industrial scouting challenge; I have filed no patent, and no contract or payment has resulted. The work received no external funding.

Author affiliation and email address. I am an independent researcher with no current institutional affiliation, and I am the sole author. The corresponding-author email address, sandler.leon@gmail.com, is my personal address and is the one registered in the submission system.

A generative AI tool (Claude, Anthropic) assisted with code development, analysis, drafting and editing; I reviewed and checked all of it and take full responsibility for the content, as disclosed in the manuscript. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author and have no competing interests.

Thank you for considering this submission.

Sincerely,

Leon Sandler